Cubic Invariant Tail Descent

All-degree harmonic identities, exact digamma cubes,
polylogarithmic representations, and algebraic Stieltjes traces

Research continuation prepared for Vladimir Reshetnikov
ProveIt / Analysis / Polylogarithms
Mathematical derivations and computational checks with OpenAI assistance

10 October 2026

Abstract

The first mixed Rogers–Dougall resonance already present in the ProveIt manuscript has a symmetric cubic structure that is not visible in its original two-parameter coefficient tables. We expose that structure and prove an all-degree descent theorem: the weighted, symmetrized product of $d \geq 2$ odd zeta tails has an explicit expression in multiple zeta values of depth at most $\max(1, d-2)$. The weight is preserved after the natural shift by two. A finite recurrence supplies every coefficient.

The first consequences include

$$\sum_{n\geq 0}(2n+1)\big(\zeta(3)-H_n^{(3)}\big)^3 = \frac{7}{4}\zeta(7),$$

and

$$\sum_{n\geq 0}(2n+1)\big(\zeta(3)-H_n^{(3)}\big)^4 = \zeta(10)+\zeta(5)^2+2\zeta(7,3).$$

We also obtain meromorphic closed formulas for the square and cube of a symmetric digamma tail, an exact cubic identity involving H_n, H_{2n} and $\log 2$, and convergent polylogarithmic integral and primitive formulas. The same cubic coordinates give a branch-independent expansion of Hurwitz and Lerch traces, all-index generalized Stieltjes identities, and anchored antiderivatives of every positive integer order. Root collisions cause no singularity in these trace formulas.

The proofs are analytic and algebraic, not integer-relation inferences. The accompanying finite exact checks and numerical diagnostics are separate from the proofs. The classical Gamma and tail-product inputs are attributed; no comprehensive historical-priority claim is made. The Gaussian S_6 and revised S_8 conjectures remain unresolved by this article.

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1 Research position and principal results

The relevant canonical starting point is `chapters/07-mixed-dougall-identities.tex` [1]. It proves a normally convergent first-resonance Gamma identity, mixed parameter derivatives, a weighted trigamma-difference square, and an all-order quadratic zeta-tail reduction. Those results are inputs here, not new claims. The harmonic coefficient chapter separately proves the Gauss identities that resolve and correct the indicated Choi formulas [2]. We do not reopen those resolved conjectures.

The contribution of this continuation is a precise structural consequence of the mixed resonance. Introducing the elementary invariants of the three roots $p, q, -p-q$ turns a large family of mixed derivatives into a triangular system ordered by the number of tail factors. Its leading term is an arbitrary symmetric convolution of odd tails; every other term has at least two fewer factors. This gives an explicit all-degree depth reduction, rather than another unorganized list of derivatives.

Let

$$T_s(n) = \zeta(s, n+1) = \zeta(s) - H_n^{(s)}, \qquad H_n^{(s)} = \sum_{j=1}^{n} j^{-s}, \qquad n \geq 0, \tag{1.1}$$

initially with integer $s \geq 3$. In particular $H_0^{(s)} = 0$. For $a \geq 0$ and $d \geq 2$, set

$$M_{a,d} = \sum_{n\geq 0} (2n+1) \sum_{\substack{j_1+\cdots+j_d=a \\ j_i \geq 0}} \prod_{i=1}^{d} T_{2j_i+3}(n). \tag{1.2}$$

The index tuples in the inner sum are ordered. This multiplicity is important: $M_{1,3} = 3\sum_n (2n+1) T_3(n)^2 T_5(n)$.

Theorem 1.1 (Summary of the tail descent)**.** *For every $a \geq 0, d \geq 2$, $M_{a,d}$ has an explicit rational linear expression in convergent multiple zeta values of weight $2a+3d-2$ and depth at most $\max(1, d-2)$. The construction is the finite recurrence in Theorem 4.2 followed by the ordered-partition map in Lemma 5.1. In particular,*

$$M_{a,2} = \frac{2a+3}{2}\zeta(2a+4), \tag{1.3}$$

$$M_{a,3} = \frac{(a+1)(2a+7)}{4}\zeta(2a+7). \tag{1.4}$$

Here and below "depth at most" is a constructive upper bound on a representation. It is not a minimal-depth assertion, a statement of period independence, or a proof that $\zeta(7,3)$ cannot be reduced further.

For a second accessible family, define

$$R_n = 2\big(\log 2 + H_n - H_{2n}\big) - \frac{1}{2n+1}. \tag{1.5}$$

We prove

$$\sum_{n\geq 0} (2n+1) R_n^2 = \frac{\zeta(2)-1}{4}, \qquad \sum_{n\geq 0} (2n+1) R_n^3 = \frac{5}{8}\zeta(3) - \log 2. \tag{1.6}$$

These are ordinary absolutely convergent series. Their proof gives a full complex-parameter identity, not just these two evaluations.

1.1 Classical antecedents and novelty boundaries

The Rogers–Dougall very well-poised summation is classical [4]. The passage from products of zeta tails to multiple zeta values is also established; Hoffman's treatment is a direct antecedent [7]. Our minimum-square version of that passage is proved below because the weight $2n+1$ and the inclusion of $n=0$ change the normalization. It is not presented as the discovery of the general sums-of-tails method.

The symmetric Gamma product, the Hurwitz shift expansion, and the Gamma–zeta derivative at zero are standard analytic ingredients [5, 6]. The additional results are their explicit cubic invariant organization, the associated all-degree descent, the closed symmetric-digamma cube, and the compatible trace/primitive formulas. These statements are proved here. A targeted search did not establish comprehensive historical priority for their individual special cases.

2 The cubic Gamma quotient

Let

$$\Omega = \{(\sigma,\tau) \in \mathbb{C}^2 : |\sigma| + |\tau| < 1\}. \tag{2.1}$$

Write r_1, r_2, r_3 for the roots, with multiplicity, of

$$r^3 - \sigma r + \tau = 0. \tag{2.2}$$

Rouché's theorem on $|r| = 1$ gives $|r_j| < 1$ on Ω. The symmetric expressions below do not require a continuous labeling of these roots. Put

$$Q_n(\sigma,\tau) = \prod_{j=1}^{3} \frac{\Gamma(n+1+r_j)}{\Gamma(n+1-r_j)}, \qquad x_n = n + \tfrac{1}{2}, \tag{2.3}$$

$$B(\sigma,\tau) = \psi(\tfrac{1}{2}) + \frac{1}{2}\left\{\psi(2) - \sum_{j=1}^{3} \psi(1+r_j)\right\}. \tag{2.4}$$

Lemma 2.1 (Product and logarithm)**.** *On Ω,*

$$Q_n(\sigma,\tau) = \prod_{j>n} \frac{j^3 - \sigma j + \tau}{j^3 - \sigma j - \tau}, \tag{2.5}$$

and the logarithm that is zero at $(0,0)$ has the locally normally convergent expansion

$$\log Q_n = 2 \sum_{a,b\geq 0} \frac{\binom{a+2b}{a}}{2b+1} T_{2a+6b+3}(n)\, \sigma^a \tau^{2b+1}. \tag{2.6}$$

Moreover,

$$x_n(Q_n - 1) - \frac{\tau}{x_n} = O(n^{-3}) \tag{2.7}$$

uniformly on compact subsets of Ω.

Proof. The relations among the roots are $\sum r_j = 0$, $\sum_{i<j} r_i r_j = -\sigma$ and $r_1 r_2 r_3 = -\tau$. The linear exponential factors in the Weierstrass product for Gamma cancel. The remaining product gives

(2.5). Equivalently, use the Gamma recurrence to remove its first n factors. Both numerator and denominator in (2.5) are nonzero on Ω.

For $j \geq 1$, expand

$$\log \frac{1+\tau/(j^3-\sigma j)}{1-\tau/(j^3-\sigma j)} = 2\sum_{b\geq 0} \frac{\tau^{2b+1}}{(2b+1)(j^3-\sigma j)^{2b+1}},$$

then expand $(1-\sigma/j^2)^{-2b-1}$. On every closed smaller domain $|\sigma|+|\tau| \leq \rho < 1$, these geometric expansions admit an absolute majorant. Summation over $j > n$ proves (2.6) and its normal convergence.

The first term is $2\tau T_3(n)$. All other terms are $O(n^{-4})$ uniformly on compact parameter sets. The elementary centered tail expansion is

$$T_3(n) = \frac{1}{2x_n^2} + O(x_n^{-4}).$$

For example, it follows from the first two terms of Euler–Maclaurin in $n+1$ followed by the substitution $n+1 = x_n + 1/2$; the cubic inverse power cancels. Thus $\log Q_n = \tau/x_n^2 + O(x_n^{-4})$ and exponentiation proves (2.7). □

Theorem 2.2 (Cubic invariant master identity)**.** *For $(\sigma,\tau) \in \Omega$,*

$$\sum_{n\geq 0} \left\{ x_n(Q_n(\sigma,\tau)-1) - \frac{\tau}{x_n} \right\} = \tau B(\sigma,\tau). \tag{2.8}$$

The bracketed sum is locally normally convergent. Every fixed derivative in σ,τ may be taken termwise, provided the displayed subtraction is retained before differentiation.

This theorem is the $a = 1$ specialization of the canonical mixed resonance, rewritten symmetrically. To make the present continuation independently readable, the next section supplies a direct derivation from the classical summation, including the finite constant. It is not a second novelty claim for that resonance.

3 A proof of the master identity with its subtraction visible

Initially choose small negative real p, q and then a sufficiently small complex neighborhood of a generic such pair. Put $d_0 = -p-q$, with $\operatorname{Re} d_0 > 0$, and set

$$\sigma = p^2+pq+q^2, \qquad \tau = pq(p+q).$$

Then the three roots are $p, q, -p-q$. Introduce a separate spectral parameter u and define

$$r_n(u) = x_n \frac{\Gamma(n+1+p)\Gamma(n+1+q)\Gamma(n-p-q-u)}{\Gamma(n+1-p)\Gamma(n+1-q)\Gamma(n+2+p+q+u)}. \tag{3.1}$$

For $0 < \operatorname{Re} u < \operatorname{Re} d_0$, the Rogers–Dougall summation, in the normalization of (3.1), is

$$\sum_{n\geq 0} r_n(u) = G(u) := \frac{\Gamma(1+p)\Gamma(1+q)\Gamma(-p-q-u)\Gamma(u)}{2\Gamma(-p-q)\Gamma(1+p+u)\Gamma(1+q+u)}. \tag{3.2}$$

Indeed substitute the upper parameters $1, 3/2, 1+p, 1+q, -p-q-u$ in the classical ${}_5F_4(1)$ formula [4, Eq. 16.4.9]; its excess is $2u$. The factor $1/2$ comes from $x_n = n + 1/2$.

The Gamma ratio in (3.1) is centered at x_n. The numerator shifts, including the canceled pair $\Gamma(n+1)/\Gamma(n+1)$, are

$$\alpha = (\tfrac{1}{2}, \tfrac{1}{2}+p, \tfrac{1}{2}+q, -\tfrac{1}{2}-p-q-u),$$

and the denominator shifts are $1-\alpha_j$. The shifted logarithmic Gamma expansion and $B_k(1-z) = (-1)^k B_k(z)$ give

$$r_n(u) = x_n^{-1-2u}\{1 + C_1(u)x_n^{-2} + O(x_n^{-4})\}, \qquad C_1(u) = -\frac{1}{3}\sum_{j=1}^{4} B_3(\alpha_j). \tag{3.3}$$

The odd inverse powers cancel before exponentiation. These estimates are uniform on compact parameter sets avoiding the displayed Gamma poles [5]. Consequently

$$\mathcal{R}(u) = \sum_{n\geq 0}\{r_n(u) - x_n^{-1-2u} - C_1(u)x_n^{-3-2u}\} \tag{3.4}$$

$$= G(u) - \zeta(1+2u, \tfrac{1}{2}) - C_1(u)\zeta(3+2u, \tfrac{1}{2}) \tag{3.5}$$

continues holomorphically across $u = -1$. Locally the summands are $O(n^{-5-2\operatorname{Re} u})$, so the continuation is also the normally convergent subtracted series there.

Write $v = u + 1$. Direct Bernoulli algebra gives

$$C_1(-1) = \tau, \qquad \frac{1}{2}C_1'(-1) + \zeta(-1, \tfrac{1}{2}) = \frac{(p+q)^2}{2}. \tag{3.6}$$

For clarity, $B_3(1/2+t) = t^3 - t/4$, and $C_1'(-1) = B_2(1/2-p-q) = (p+q)^2 - 1/12$; $\zeta(-1, 1/2) = 1/24$.

The Laurent constant of $G(-1+v)$ is

$$G(-1+v) = \frac{\tau}{2v} + \frac{\tau}{2}\{\psi(2) - \psi(1-p-q) - \psi(p) - \psi(q)\} + O(v). \tag{3.7}$$

The resonant Hurwitz term is

$$C_1(-1+v)\zeta(1+2v, \tfrac{1}{2}) = \frac{\tau}{2v} + \frac{1}{2}C_1'(-1) - \tau\psi(\tfrac{1}{2}) + O(v).$$

Subtracting these expansions in (3.5), and using $\psi(p) = \psi(1+p) - 1/p$ and its q counterpart, cancels precisely

$$\frac{\tau}{2}\left(\frac{1}{p} + \frac{1}{q}\right) = \frac{(p+q)^2}{2}.$$

Since $r_n(-1) = x_n Q_n$, the result is (2.8).

The initial restrictions on p, q can now be removed by holomorphy. Both sides are symmetric in the three roots. Alternatively, regard them as holomorphic functions of σ, τ using the product and power-series representations. The map $(p, q) \mapsto (\sigma, \tau)$ has nonzero Jacobian at generic small pairs, so the identity holds on a nonempty open subset of Ω. The identity theorem and connectedness of Ω extend it throughout. Lemma 2.1 gives normal convergence directly on this entire domain. Cauchy's integral formula on a slightly larger compact polydisc then justifies all fixed coefficient extractions and derivatives. □

4 Newton coefficients and the all-degree tail recurrence

Put $S_k = r_1^k + r_2^k + r_3^k$. The relation

$$-\log(1 - \sigma t^2 + \tau t^3) = \sum_{k\geq 1} \frac{S_k}{k} t^k \tag{4.1}$$

gives, for $k \geq 1$,

$$S_k = \sum_{\substack{2a+3b=k \\ a,b\geq 0}} \frac{k}{a+b} \binom{a+b}{b} (-1)^b \sigma^a \tau^b. \tag{4.2}$$

There is no $(a, b) = (0, 0)$ term. Equivalently, $S_0 = 3, S_1 = 0, S_2 = 2\sigma$ and $S_k = \sigma S_{k-2} - \tau S_{k-3}$.

Using $\psi(1 + z) = -\gamma + \sum_{k\geq 1}(-1)^{k+1}\zeta(k + 1)z^k$, the two signs $(-1)^k$ and $(-1)^b$ cancel when $k = 2a + 3b$. Hence

$$B(\sigma, \tau) = \frac{1}{2} - 2\log 2 + \frac{1}{2} \sum_{\substack{a,b\geq 0 \\ a+b>0}} \frac{2a + 3b}{a + b} \binom{a+b}{b} \zeta(2a + 3b + 1)\sigma^a \tau^b. \tag{4.3}$$

The expansion is normally convergent on smaller polydiscs inside Ω. The power-sum formula, or the root trace itself, continues it wherever needed in Ω.

Definition 4.1. Treat $T_3, T_5, \ldots$ as independent formal symbols. Define

$$P_{a,d} = [\sigma^a \tau^d] \exp\left(2 \sum_{i,b\geq 0} \frac{\binom{i+2b}{i}}{2b + 1} T_{2i+6b+3} \sigma^i \tau^{2b+1} \right). \tag{4.4}$$

This is a finite polynomial for each fixed (a, d).

Theorem 4.2 (Finite recurrence and exact summation)**.** *The initial values are $P_{0,0} = 1$ and $P_{a,0} = 0$ for $a > 0$. For $d \geq 1$,*

$$P_{a,d} = \frac{2}{d} \sum_{\substack{b\leq d \\ b \text{ odd}}} \sum_{i=0}^{a} \binom{i + b - 1}{i} T_{2i+3b} P_{a-i,d-b}. \tag{4.5}$$

For every $a \geq 0, d \geq 2$,

$$\sum_{n\geq 0} (2n + 1) P_{a,d}(T_3(n), T_5(n), \ldots) = C_{a,d}\zeta(2a + 3d - 2), \tag{4.6}$$

where

$$C_{a,d} = \frac{2a + 3d - 3}{a + d - 1} \binom{a + d - 1}{d - 1}. \tag{4.7}$$

All the sums in (4.6) *converge absolutely.*

Proof. Differentiating the exponential in (4.4) with respect to τ gives (4.5). On substituting the genuine tails, $P_{a,d}$ is exactly $[\sigma^a \tau^d]Q_n$. Take this coefficient in Theorem 2.2 and multiply by two. When $d \geq 2$, the counterterm and the constant term of B make no contribution. The coefficient in (4.3) with $b = d - 1$ gives (4.7).

Each monomial is a product of $\ell \geq 1$ tails, all of orders at least three, with total order $2a + 3d$. Because $T_s(n) = O(n^{1-s})$, its weighted summand is $O(n^{1+\ell-2a-3d})$. Here $\ell \leq d$, so the exponent is at most $1 - 2a - 2d \leq -3$. This also directly proves absolute convergence without relying only on coefficient extraction. □

4.1 The exceptional linear coefficient

The sole coefficient requiring the subtraction is $(a, d) = (0, 1)$:

$$\sum_{n\geq 0}\left\{(2n+1)T_3(n) - \frac{1}{n+1/2}\right\} = \frac{1}{2} - 2\log 2. \tag{4.8}$$

For $a \geq 1, d = 1$, the identity is simply $\sum_n (2n+1)T_{2a+3}(n) = \zeta(2a+1)$. Equation (4.8) must not be split into the difference of two ordinarily convergent sums: both separate series diverge. No such finite-part issue is present in the nonlinear identities.

5 Why two levels of depth disappear

Use the convention

$$\zeta(s_1, \ldots, s_\ell) = \sum_{m_1 > \cdots > m_\ell \geq 1} m_1^{-s_1} \cdots m_\ell^{-s_\ell}. \tag{5.1}$$

An ordered set partition of $\{1, \ldots, d\}$ is a list of nonempty disjoint blocks whose union is that set; the order of the blocks matters.

Lemma 5.1 (The minimum-square map)**.** *For integers $s_i \geq 3$,*

$$\sum_{n\geq 0}(2n+1)\prod_{i=1}^{d} T_{s_i}(n) = \sum_{(B_1,\ldots,B_\ell)} \zeta\left(\sum_{i\in B_1} s_i, \ldots, \sum_{i\in B_\ell} s_i - 2\right), \tag{5.2}$$

where the last exponent alone is reduced by two. A one-block partition contributes $\zeta(s_1 + \cdots + s_d - 2)$.

Proof. Expand the product as a sum over $m_i > n$. Positivity permits Tonelli, and

$$\sum_{0\leq n<\min(m_1,\ldots,m_d)} (2n+1) = \min(m_1, \ldots, m_d)^2.$$

Group the indices m_i according to their equal values, listing the blocks from largest value to smallest. The minimum-square factor reduces the last exponent by two and produces exactly the summand in (5.2). No factorial is inserted: every ordered partition already records one distinct equality-and-order pattern. All resulting multiple zeta series converge. Their first exponent is at least three unless there is just one block, and their final exponent is at least one. Alternatively, the preceding tail estimate proves the finiteness of the original positive sum. □

This is the weighted counterpart of the classical tail-product rearrangement discussed by Hoffman [7].

Define

$$A_n(\sigma) = \sum_{j\geq 0} T_{2j+3}(n)\sigma^j.$$

The part of $\log Q_n$ linear in τ is $2\tau A_n(\sigma)$. Every remaining term has odd τ-degree at least three. Consequently

$$P_{a,d} = \frac{2^d}{d!}[\sigma^a]\left(\sum_{j\geq 0} T_{2j+3}\sigma^j\right)^d + E_{a,d}, \tag{5.3}$$

where every monomial of $E_{a,d}$ has at most $d-2$ tail factors and the same parity of factor count as d. This is the entire reason for the depth drop; it is not a numerical rank observation.

Theorem 5.2 (Constructive depth descent)**.** *For $a \geq 0, d \geq 2$,*

$$M_{a,d} = \frac{d!}{2^d}\left\{ C_{a,d}\zeta(2a+3d-2) - \sum_{n\geq 0}(2n+1)E_{a,d}(T_3(n), T_5(n), \ldots)\right\}. \tag{5.4}$$

Applying (5.2) *to the finitely many monomials in $E_{a,d}$ produces only multiple zeta values of weight $2a+3d-2$ and depth at most $d-2$. For $d=2$, $E_{a,2}=0$, so the result has depth one.*

Proof. The unique way to obtain d tail factors at τ-degree d in the exponential is to use d linear terms. If any factor has τ-degree at least three, the number of factors decreases by at least two. This proves (5.3). Substitution in (4.6) proves (5.4). Each monomial has total tail order $2a+3d$; Lemma 5.1 reduces that weight by exactly two and never increases its number of factors. All operations involve finite linear combinations of absolutely convergent series. □

The theorem concerns the symmetrized convolution $M_{a,d}$, not every individual product with the same weight. For instance it directly reduces the sum of $T_3^2T_7$ and $T_3T_5^2$, but does not separately evaluate both. That distinction prevents an unjustified polarization claim.

6 Cubes, fourth powers, and a fifth-power relation

6.1 The complete quadratic and cubic families

The τ^2 coefficient is $2A_n(\sigma)^2$, giving (1.3). At degree three, put

$$C_n(\sigma) = \sum_{a\geq 0}\binom{a+2}{2}T_{2a+9}(n)\sigma^a.$$

Then

$$[\tau^3]Q_n = \frac{4}{3}A_n(\sigma)^3 + \frac{2}{3}C_n(\sigma). \tag{6.1}$$

The elementary identity $\sum_n(2n+1)T_s(n) = \zeta(s-2)$ for $s>3$ implies

$$\frac{4}{3}M_{a,3} + \frac{2}{3}\binom{a+2}{2}\zeta(2a+7) = (a+1)(a+3)\zeta(2a+7).$$

Solving proves (1.4). Explicit consequences are

$$\sum_n(2n+1)T_3(n)^3 = \frac{7}{4}\zeta(7), \tag{6.2}$$

$$\sum_n(2n+1)T_3(n)^2T_5(n) = \frac{3}{2}\zeta(9), \tag{6.3}$$

$$\sum_n(2n+1)\{T_3(n)^2T_7(n) + T_3(n)T_5(n)^2\} = \frac{11}{4}\zeta(11). \tag{6.4}$$

For example, the first identity also states

$$6\zeta(3,3,1) + 3\zeta(3,4) + 3\zeta(6,1) + \zeta(7) = \frac{7}{4}\zeta(7). \tag{6.5}$$

This is an equality of periods with a proof from the master identity; it is not an assertion that the four displayed coordinates form a basis.

6.2 A fourth power reduces to one depth-two coordinate

At $a = 0$,

$$P_{0,4} = \frac{2}{3}T_3^4 + \frac{4}{3}T_3T_9. \tag{6.6}$$

Since $C_{0,4} = 3$, Theorem 4.2 gives

$$\sum_n (2n+1)T_3(n)^4 = \frac{9}{2}\zeta(10) - 2\sum_n (2n+1)T_3(n)T_9(n). \tag{6.7}$$

The minimum-square map yields

$$\sum_n (2n+1)T_3(n)T_9(n) = \zeta(10) + \zeta(3,7) + \zeta(9,1).$$

Euler's depth-two reduction and the ordinary stuffle product give

$$\begin{aligned}\zeta(9,1) &= \frac{9}{2}\zeta(10) - \zeta(2)\zeta(8) - \zeta(3)\zeta(7) - \zeta(4)\zeta(6) - \tfrac{1}{2}\zeta(5)^2,\\ \zeta(3,7) + \zeta(7,3) &= \zeta(3)\zeta(7) - \zeta(10).\end{aligned}$$

These classical relations are recorded, with the same series convention, in [7, §2]. Substituting them and using $\zeta(2)\zeta(8) = 33\zeta(10)/20$ and $\zeta(4)\zeta(6) = 11\zeta(10)/10$ proves the following.

Corollary 6.1.

$$\sum_{n\geq 0} (2n+1)\big(\zeta(3) - H_n^{(3)}\big)^4 = \zeta(10) + \zeta(5)^2 + 2\zeta(7,3). \tag{6.8}$$

6.3 The next descent, without claiming a complete basis reduction

The fifth coefficient is

$$P_{0,5} = \frac{4}{15}T_3^5 + \frac{4}{3}T_3^2T_9 + \frac{2}{5}T_{15}.$$

It follows that

$$\sum_n (2n+1)\{T_3(n)^5 + 5T_3(n)^2T_9(n)\} = \frac{39}{4}\zeta(13). \tag{6.9}$$

For a fully explicit depth-three form, the remaining mixed moment is

$$\begin{aligned}\sum_n (2n+1)T_3(n)^2T_9(n) &= 2\zeta(3,3,7) + 2\zeta(3,9,1) + 2\zeta(9,3,1)\\ &\quad + \zeta(6,7) + \zeta(9,4) + 2\zeta(12,1) + 2\zeta(3,10) + \zeta(13).\end{aligned} \tag{6.10}$$

The coefficients count the ordered partitions of the two labeled 3-tails and one labeled 9-tail. Thus (6.9) really is a finite reduction of the fifth power to depth at most three. No numerical basis selection is needed to make it a proved exact identity.

7 Closed meromorphic formulas for symmetric digamma powers

For $x \notin \mathbb{Z} \setminus \{0\}$, define

$$\mathcal{A}_n(x) = \sum_{j>n} \frac{1}{j(j^2 - x^2)} = \frac{2\psi(n+1) - \psi(n+1-x) - \psi(n+1+x)}{2x^2}, \tag{7.1}$$

with $\mathcal{A}_n(0) = T_3(n)$. The equality follows by partial fractions and the digamma difference series. For $|x| < 1$, $\mathcal{A}_n(x) = A_n(x^2)$.

Set

$$J(x) = \psi(1+x) + \psi(1-x) - 2\psi(1). \tag{7.2}$$

Then $J'(x) = \psi'(1+x) - \psi'(1-x)$ and $J''(x) = \psi''(1+x) + \psi''(1-x)$.

Theorem 7.1 (Exact square and cube)**.** *On* $\mathbb{C} \setminus (\mathbb{Z} \setminus \{0\})$,

$$\sum_{n\geq 0}(2n+1)\mathcal{A}_n(x)^2 = \frac{\pi^2}{4x^2\sin^2(\pi x)} - \frac{\zeta(2)}{2x^2} - \frac{1}{4x^4}, \tag{7.3}$$

$$\sum_{n\geq 0}(2n+1)\mathcal{A}_n(x)^3 = \frac{J(x)}{4x^6} + \frac{J'(x)}{8x^5} - \frac{J''(x)}{16x^4} + \frac{3\zeta(3)}{4x^4}. \tag{7.4}$$

All singularities at $x = 0$ on the right are removable. The powers are algebraic powers, not absolute-value powers. The series converge locally normally away from the nonzero integers.

Proof. First let $\sigma = x^2$, $|x| < 1$, and $D = \sigma\partial_\sigma$. The quadratic family gives

$$\sum_n (2n+1)\mathcal{A}_n(x)^2 = (D + \tfrac{3}{2})F(\sigma), \qquad F(\sigma) = \sum_{a\geq 0}\zeta(2a+4)\sigma^a.$$

Using the partial-fraction expansion of cotangent,

$$E(\sigma) = \sum_{j\geq 1}\frac{1}{j^2-\sigma} = \frac{1-\pi x\cot(\pi x)}{2x^2}, \qquad F(\sigma) = \frac{E(\sigma)-\zeta(2)}{\sigma}.$$

Differentiation cancels the cotangent terms and gives (7.3).

Similarly, the cubic family yields

$$\sum_n (2n+1)\mathcal{A}_n(x)^3 = \tfrac{1}{4}(D+1)(2D+7)G(\sigma),$$

$$G(\sigma) = \sum_{a\geq 0}\zeta(2a+7)\sigma^a = \frac{\mathcal{A}_0(x) - \zeta(3) - x^2\zeta(5)}{x^4}.$$

Since $\mathcal{A}_0(x) = -J(x)/(2x^2)$ and $D = (x/2)\partial_x$, this operator is $x^2\partial_x^2/8 + 5x\partial_x/4 + 7/4$. Its action on $-J(x)/(2x^6) - \zeta(3)/x^4 - \zeta(5)/x^2$ gives (7.4); the $\zeta(5)$ term cancels identically.

On a compact set avoiding the nonzero integers, the finitely many small j denominators are harmless and the large-n tail satisfies $\mathcal{A}_n(x) = O(n^{-2})$ uniformly. Thus the squared and cubed series are normally convergent there and define meromorphic functions of x. The identity theorem extends the formulas from the unit disc to the connected punctured plane. Their removable values at zero agree with (1.3) and (6.2). □

7.1 A half-shift harmonic cube

The half-integer recurrence gives

$$\mathcal{A}_n(\tfrac{1}{2}) = 8(\log 2 + H_n - H_{2n}) - \frac{4}{2n+1} = 4R_n.$$

At $x = 1/2$,

$$J = 2 - 4\log 2, \qquad J' = -4, \qquad J'' = 16 - 28\zeta(3).$$

Equations (7.3)–(7.4) therefore give

$$\sum_n (2n+1)\mathcal{A}_n(\tfrac{1}{2})^2 = 4\zeta(2) - 4, \qquad \sum_n (2n+1)\mathcal{A}_n(\tfrac{1}{2})^3 = 40\zeta(3) - 64\log 2.$$

Division by 16 and 64, respectively, proves (1.6).

There is also an elementary integral realization,

$$R_n = \int_0^1 t^{2n} \frac{1-t}{1+t}\,\mathrm{d}t. \tag{7.5}$$

Indeed $\int_0^1 t^{2n}/(1+t)\,\mathrm{d}t = \log 2 + H_n - H_{2n}$. Thus one further exact integral consequence is

$$\int_{[0,1]^3} \frac{1+(xyz)^2}{(1-(xyz)^2)^2} \frac{1-x}{1+x}\frac{1-y}{1+y}\frac{1-z}{1+z}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \frac{5}{8}\zeta(3) - \log 2. \tag{7.6}$$

Tonelli applies to the positive power series $\sum_{n\geq 0}(2n+1)(xyz)^{2n}$; its integral is finite by the already proved cubic moment. There is no hidden endpoint finite part.

7.2 A colliding-root Gamma example

Taking $p = q = 1/4$ gives $\sigma = 3/16, \tau = 1/32$. The cubic has a repeated root, but the master identity is regular and yields

$$\sum_{n\geq 0}\left[(n+\tfrac{1}{2})\left\{\frac{\Gamma(n+5/4)^2\Gamma(n+1/2)}{\Gamma(n+3/4)^2\Gamma(n+3/2)} - 1\right\} - \frac{1}{32(n+1/2)}\right] = \frac{\pi + 4\log 2 - 7}{64}. \tag{7.7}$$

This example illustrates why singular derivatives of individually labeled roots should not be confused with singularities of a symmetric trace.

8 Polylogarithmic representations and explicit primitives

For positive integers $s_1, \ldots, s_\ell$ and $|z| < 1$, use

$$\mathrm{Li}_{s_1,\ldots,s_\ell}(z) = \sum_{m_1 > \cdots > m_\ell \geq 1} \frac{z^{m_1}}{m_1^{s_1}\cdots m_\ell^{s_\ell}}. \tag{8.1}$$

Only the largest summation index carries z. This convention fixes the primitive formulas and the endpoint $\mathrm{Li}_{\boldsymbol{s}}(1) = \zeta(\boldsymbol{s})$.

Proposition 8.1 (An ordinary polylogarithmic realization of the fourth power)**.**

$$\sum_{n\geq 0}(2n+1)T_3(n)^4 = \zeta(10) + \zeta(5)^2 + \frac{1}{360}\int_0^1 \frac{(-\log t)^6\,\mathrm{Li}_3(t)}{1-t}\,\mathrm{d}t. \tag{8.2}$$

Proof. The geometric series and the defining series of Li_3 give

$$\frac{1}{6!}\int_0^1 \frac{(-\log t)^6\,\mathrm{Li}_3(t)}{1-t}\,\mathrm{d}t = \sum_{m>k\geq 1}\frac{1}{m^7k^3} = \zeta(7,3).$$

All terms are positive. The integrand is integrable: near zero the factor $\mathrm{Li}_3(t)$ is $O(t)$, and near one the sixth logarithmic power compensates the denominator. Apply Corollary 6.1. □

Proposition 8.2 (An explicit primitive ladder)**.** *For $m \geq 0$, $q \geq 1$ an integer, and $0 < z < 1$,*

$$\int_0^z \frac{(-\log t)^m\,\mathrm{Li}_q(t)}{1-t}\,\mathrm{d}t = \sum_{j=0}^{m}\frac{m!}{(m-j)!}(-\log z)^{m-j}\,\mathrm{Li}_{j+1,q}(z). \tag{8.3}$$

On $|z| < 1$, the same identity holds with consistently chosen logarithms and paths from zero. It can also be differentiated any fixed number of times in the order q when Li_q and the nested series are regarded as entire functions of that order in the open disc.

Proof. The defining series give

$$\frac{\mathrm{d}}{\mathrm{d}z}\,\mathrm{Li}_{1,q}(z) = \frac{\mathrm{Li}_q(z)}{1-z}, \qquad \frac{\mathrm{d}}{\mathrm{d}z}\,\mathrm{Li}_{r,q}(z) = \frac{\mathrm{Li}_{r-1,q}(z)}{z} \quad (r>1).$$

Differentiating the finite sum on the right of (8.3) telescopes; only the desired integrand remains. Every term tends to zero as $z \to 0$ along the chosen path because its nested series starts at order z^2. This fixes the integration constant. Local uniform geometric convergence, including every fixed order derivative, justifies the last assertion. At $z = 1$, the endpoint limit for $m \geq 1$ is $m!\zeta(m+1,q)$ whenever the defining endpoint series converges. □

These integration rules are standard consequences of the iterated-series calculus; their role here is to turn the depth-descent output into ordinary polylogarithmic integrals and explicit antiderivatives.

8.1 Generating functions for arbitrary powers

For $s \geq 3$, define

$$F_{s,d}(z) = \sum_{n\geq 0} T_s(n)^d z^n, \qquad |z| < 1.$$

For $k \geq 1$ put

$$V_{s,k}(z) = \sum_{\ell=1}^{k}\sum_{\substack{m_1+\cdots+m_\ell=k\\ m_i\geq 1}}\frac{k!}{m_1!\cdots m_\ell!}\,\mathrm{Li}_{sm_1,\ldots,sm_\ell}(z), \qquad V_{s,0}(z) = 1.$$

Expanding the kth power of $H_n^{(s)}$ by the equal-index blocks gives

$$F_{s,d}(z) = \frac{1}{1-z}\sum_{k=0}^{d}(-1)^k\binom{d}{k}\zeta(s)^{d-k}V_{s,k}(z). \tag{8.4}$$

There is no appeal to numerical multiple-polylogarithm evaluation in this identity. The coefficient of z^n is exactly $(\zeta(s) - H_n^{(s)})^d$.

Writing $\theta = z\partial_z$,

$$(2\theta+1)F_{s,d}(z) = \sum_{n\geq 0}(2n+1)T_s(n)^d z^n. \tag{8.5}$$

For $s = 3, d \geq 2$, absolute convergence permits $z \uparrow 1$ on the right. The cubic and quartic evaluations above are therefore explicit endpoint cancellation identities for the multiple polylogarithms in (8.4). One must take the complete combination on the left; its individual rationally weighted pieces need not have finite endpoint limits.

9 Algebraic Hurwitz traces and all-index Stieltjes identities

We now use the same cubic roots, but allow a positive base point h with $h > \max_j |r_j|$. This condition ensures that every $h + r_j$ lies in the right half-plane. Let

$$\mathcal{Z}(s;h) = \sum_{j=1}^{3}\zeta(s,h+r_j) - 3\zeta(s,h). \tag{9.1}$$

The residues at $s = 1$ cancel, so $\mathcal{Z}$ is entire in s. The dependence on σ, τ is symmetric and holomorphic, including at the cubic discriminant $4\sigma^3 - 27\tau^2 = 0$.

Theorem 9.1 (Invariant spectral trace expansion)**.** *For $h > \max_j |r_j|$,*

$$\mathcal{Z}(s;h) = \sum_{\substack{a,b\geq 0\\ a+b>0}} \frac{k}{a+b}\binom{a+b}{b}\frac{(s)_k}{k!}\zeta(s+k,h)\sigma^a\tau^b, \qquad k = 2a+3b. \tag{9.2}$$

The grouped series is locally normally convergent in s and in parameter sets on which all roots remain strictly inside $|r| < h$. An explicit sufficient polydisc is $|\sigma|/h^2 + |\tau|/h^3 < 1$. Every apparent pole of an individual spectral factor at $s = 1-k$ is removable.

Proof. The ordinary Taylor expansion in the Hurwitz shift is

$$\zeta(s,h+r) = \sum_{k\geq 0}\frac{(-1)^k(s)_k}{k!}\zeta(s+k,h)r^k.$$

It follows initially from the absolutely convergent defining series for $\operatorname{Re} s > 1$, and then by analytic continuation. The $k = 0$ terms cancel in the trace; the $k = 1$ term vanishes because $\sum r_j = 0$. Substitute (4.2). Since $(-1)^k = (-1)^b$ when $k = 2a + 3b$, the signs cancel and give (9.2).

For uniform convergence, choose a radius ρ strictly between the largest root modulus and h. The shift Taylor expansion is uniform on a larger circle in the shift plane, on compact spectral sets after removal of the common pole. Cauchy's estimate then gives a geometric majorant for the root power sums. Each k contains only finitely many invariant monomials; on the stated smaller polydiscs their absolute expansions have the same majorant. At $s = 1-k$, the simple pole of $\zeta(s+k,h)$ is canceled by the zero of $(s)_k$, which contains the factor $s + k - 1$. □

Remark 9.2. The locally normal statement for an arbitrary root-bounded parameter set uses the grouping by $k = 2a + 3b$. Absolute convergence of the unrestricted double series is guaranteed on the explicit smaller polydisc. Keeping this distinction avoids asserting absolute convergence merely from a favorable choice of individually labeled roots.

Define generalized Stieltjes constants by

$$\zeta(1+u,h)=\frac{1}{u}+\sum_{m\geq 0}\frac{(-1)^m}{m!}\gamma_m(h)u^m,\qquad \gamma_0(h)=-\psi(h), \tag{9.3}$$

and set

$$\Delta_m(h)=\sum_{j=1}^{3}\gamma_m(h+r_j)-3\gamma_m(h).$$

For $k\geq 1$ let $E_\ell(k)$ be the elementary harmonic coefficients

$$\prod_{j=1}^{k}(1+u/j)=\sum_{\ell\geq 0}\frac{E_\ell(k)}{\ell!}u^\ell. \tag{9.4}$$

Thus $E_0=1$, $E_1=H_k$, $E_2=H_k^2-H_k^{(2)}$ and $E_3=H_k^3-3H_kH_k^{(2)}+2H_k^{(3)}$; $E_\ell(k)=0$ for $\ell>k$.

Corollary 9.3 (All generalized Stieltjes indices)**.** *In the same convergence domains,*

$$\begin{aligned}\Delta_m(h)=(-1)^m\sum_{\substack{a,b\geq 0\\ a+b>0}}\frac{k}{a+b}\binom{a+b}{b}\sigma^a\tau^b\\ \times\sum_{\ell=0}^{m}\binom{m}{\ell}E_\ell(k)\zeta^{(m-\ell)}(k+1,h),\qquad k=2a+3b.\end{aligned} \tag{9.5}$$

The superscript on ζ denotes differentiation in its spectral argument.

Proof. Use $\Delta_m=(-1)^m\partial_s^m\mathcal{Z}(s;h)|_{s=1}$. The factor $(1+u)_k/k!$ is exactly the product in (9.4). Leibniz's rule in (9.2) gives the formula, and normal convergence justifies termwise differentiation. □

For example, the first spectral derivative gives the explicit convergent identity

$$\Delta_1(h)=-\sum_{\substack{a,b\geq 0\\ a+b>0}}\frac{k}{a+b}\binom{a+b}{b}\sigma^a\tau^b\{\zeta'(k+1,h)+H_k\zeta(k+1,h)\}. \tag{9.6}$$

This is a direct link between generalized Stieltjes constants, generalized harmonic numbers, and derivatives of Hurwitz zeta in the cubic coordinates.

9.1 The same trace is the right side of the Gamma master

At $h=1$,

$$B(\sigma,\tau)=\frac{1}{2}-2\log 2+\frac{1}{2}\mathcal{Z}(1;1)=\frac{1}{2}-2\log 2+\frac{1}{2}\Delta_0(1). \tag{9.7}$$

Thus the right side of the nonlinear Gamma identity and the all-index Stieltjes calculus have exactly the same invariant coefficients. This does *not* license differentiating the integer tail orders in the nonlinear descent theorem: a separate continuously deformed summation identity would be required for that stronger conclusion.

9.2 A rational-shift check at every Stieltjes index

Take $r = (1/2, 0, -1/2)$, so $(\sigma, \tau) = (1/4, 0)$, and $h = 1$. The Hurwitz recurrence and duplication give

$$\mathcal{Z}(s; 1) = (2^{s+1} - 4)\zeta(s) - 2^s.$$

If $L = \log 2$ and $\gamma_j = \gamma_j(1)$, then

$$\Delta_m(1) = (-1)^m \left\{ \frac{4L^{m+1}}{m+1} + 4 \sum_{j=0}^{m-1} \binom{m}{j} L^{m-j}(-1)^j \gamma_j - 2L^m \right\}, \tag{9.8}$$

with an empty sum when $m = 0$. In particular $\Delta_0(1) = 4L - 2$. The pole is canceled *before* Taylor expansion. Expanding the three Hurwitz zeta functions separately at $s = 1$ without canceling their common pole is an avoidable source of incorrect constants.

10 Gamma determinants and anchored antiderivatives

Define

$$\mathcal{D}(h; \sigma, \tau) = \frac{\prod_{j=1}^{3} \Gamma(h + r_j)}{\Gamma(h)^3}, \tag{10.1}$$

with its logarithm normalized to zero at $(\sigma, \tau) = (0, 0)$. The zeta derivative at zero and the cancellation of the three common $\log(2\pi)/2$ terms imply

$$\log \mathcal{D}(h; \sigma, \tau) = \mathcal{Z}'(0; h). \tag{10.2}$$

At $h = 1$, the elementary Gamma product gives

$$\mathcal{D}(1; \sigma, \tau) = \prod_{n \geq 1} (1 - \sigma/n^2 - \tau/n^3)^{-1}, \tag{10.3}$$

$$\log \mathcal{D}(1; \sigma, \tau) = \sum_{\substack{a,b \geq 0 \\ a+b>0}} \frac{1}{a+b} \binom{a+b}{b} \zeta(2a + 3b) \sigma^a \tau^b. \tag{10.4}$$

Also $Q_n = \mathcal{D}(n+1; \sigma, \tau)/\mathcal{D}(n+1; \sigma, -\tau)$. This identifies the odd-τ Gamma quotient that caused the depth drop.

Proposition 10.1 (Derivative tower)**.** *For every $r \geq 0$,*

$$\partial_h^r \mathcal{Z}(s; h) = (-1)^r (s)_r \mathcal{Z}(s + r; h). \tag{10.5}$$

Consequently, for $r \geq 1$,

$$\partial_h^r \Delta_m(h) = (-1)^{m+r} r! \sum_{\ell=0}^{m} \binom{m}{\ell} E_\ell(r) \mathcal{Z}^{(m-\ell)}(1 + r; h). \tag{10.6}$$

Proof. Apply the standard parameter derivative $\partial_h \zeta(s, h) = -s\zeta(s+1, h)$ repeatedly to the finite trace. Differentiate the resulting identity m times in s at $s = 1$. The Taylor coefficients of $(1 + u)_r$ are $r! E_\ell(r)/\ell!$. □

We give every positive integer order of antiderivative with its integration constants fixed. Choose real $h, h_0 > \max_j |r_j|$; the formulas extend holomorphically to an appropriate simply connected right-half-plane domain. For $r \geq 1$ set

$$N_r(u; h, h_0) = \mathcal{Z}(1 - r + u; h) - \sum_{j=0}^{r-1} \frac{(h - h_0)^j}{j!} \, \partial_t^j \mathcal{Z}(1 - r + u; t)\Big|_{t=h_0}. \tag{10.7}$$

Theorem 10.2 (All anchored Stieltjes primitives)**.** *The quotient $N_r(u; h, h_0)/(1 - r + u)_r$ has a removable singularity at $u = 0$. Define*

$$\mathcal{I}_{r,m}(h; h_0) = (-1)^{m+r} m! [u^m] \frac{N_r(u; h, h_0)}{(1 - r + u)_r}. \tag{10.8}$$

Then

$$\partial_h^r \mathcal{I}_{r,m}(h; h_0) = \Delta_m(h), \qquad \partial_h^j \mathcal{I}_{r,m}(h; h_0)\Big|_{h=h_0} = 0 \quad (0 \leq j < r). \tag{10.9}$$

Equivalently,

$$\mathcal{I}_{r,m}(h; h_0) = \frac{1}{(r-1)!} \int_{h_0}^{h} (h - t)^{r-1} \Delta_m(t) \, \mathrm{d}t. \tag{10.10}$$

Proof. The negative-integer identity $\zeta(1 - r, h) = -B_r(h)/r$ shows that $\mathcal{Z}(1 - r; h)$ is a polynomial in h of degree at most $r - 2$: its leading degree-r terms cancel between the three shifts and the base, and its degree-$(r-1)$ coefficient is a multiple of $\sum r_j = 0$. For $r = 1$ it vanishes identically. Its Taylor polynomial of degree $r - 1$ around h_0 is therefore exact, so $N_r(0; h, h_0) = 0$. The denominator has exactly one zero at $u = 0$, with derivative $(-1)^{r-1}(r-1)!$. Holomorphic division proves removability.

Differentiate (10.7) r times in h. The subtracted Taylor polynomial disappears and (10.5) gives

$$\partial_h^r N_r(u; h, h_0) = (-1)^r (1 - r + u)_r \mathcal{Z}(1 + u; h).$$

After division and coefficient extraction, the sign in (10.8) gives $(-1)^m m! [u^m] \mathcal{Z}(1 + u; h) = \Delta_m(h)$. The lower derivatives vanish at $h = h_0$ by construction, including after the removable division. The usual uniqueness of the solution of the rth derivative equation with these initial data gives (10.10). □

For the first primitive this simplifies to

$$\mathcal{I}_{1,m}(h; h_0) = \frac{(-1)^{m+1}}{m+1} \{\mathcal{Z}^{(m+1)}(0; h) - \mathcal{Z}^{(m+1)}(0; h_0)\}. \tag{10.11}$$

In particular,

$$\int_{h_0}^{h} \Delta_0(t) \, \mathrm{d}t = \log \mathcal{D}(h_0; \sigma, \tau) - \log \mathcal{D}(h; \sigma, \tau). \tag{10.12}$$

This is a Gamma antiderivative with no unspecified polynomial ambiguity. The all-order formula treats that ambiguity explicitly rather than hiding it in a symbol for a negative-order polygamma function.

11 Lerch traces and polylogarithmic order derivatives

Let $|z| < 1$ and

$$\Phi(z, s, h) = \sum_{n\geq 0} \frac{z^n}{(n+h)^s}, \qquad \mathcal{L}(z, s; h) = \sum_{j=1}^{3} \Phi(z, s, h + r_j) - 3\Phi(z, s, h).$$

All powers use the logarithm in the right half-plane. The geometric factor makes these functions entire in s. Exactly the proof of Theorem 9.1 gives

$$\mathcal{L}(z, s; h) = \sum_{\substack{a,b\geq 0\\ a+b>0}} \frac{k}{a+b}\binom{a+b}{b}\frac{(s)_k}{k!}\Phi(z, s+k, h)\sigma^a\tau^b, \qquad k = 2a + 3b. \tag{11.1}$$

At $h = 1$, $z\Phi(z, s, 1) = \mathrm{Li}_s(z)$, so

$$z\sum_{j=1}^{3} \Phi(z, s, 1 + r_j) - 3\,\mathrm{Li}_s(z) = \sum_{\substack{a,b\geq 0\\ a+b>0}} \frac{k}{a+b}\binom{a+b}{b}\frac{(s)_k}{k!}\,\mathrm{Li}_{s+k}(z)\sigma^a\tau^b. \tag{11.2}$$

Every fixed spectral derivative may be taken termwise. For example,

$$\partial_s\left\{z\sum_j \Phi(z, s, 1 + r_j) - 3\,\mathrm{Li}_s(z)\right\}\Big|_{s=1} = \sum_{\substack{a,b\geq 0\\ a+b>0}} \frac{k}{a+b}\binom{a+b}{b}\sigma^a\tau^b \times \left\{\partial_v\,\mathrm{Li}_v(z)|_{v=k+1} + H_k\,\mathrm{Li}_{k+1}(z)\right\}. \tag{11.3}$$

At higher derivative orders, the bracket is replaced by $\sum_{\ell=0}^{m}\binom{m}{\ell}E_\ell(k)\partial_v^{m-\ell}\,\mathrm{Li}_v(z)|_{v=k+1}$.

11.1 The endpoint and why it is controlled

For fixed shifts with zero sum,

$$\sum_j (n + h + r_j)^{-s} - 3(n + h)^{-s} = O(n^{-\operatorname{Re} s-2})$$

locally uniformly in s, including fixed spectral derivatives with an additional logarithmic factor. Thus the trace defining series at $z = 1$ converges ordinarily for $\operatorname{Re} s > -1$. On this half-plane Abel passage from (11.1) recovers (9.2). In particular all spectral derivatives at $s = 1$ have ordinary convergent endpoint traces. For $\operatorname{Re} s \leq -1$, analytic continuation is available but ordinary endpoint convergence is not asserted.

11.2 A finite differential system in the argument

The traces with powers of the roots also remain finite-dimensional. Put

$$F_j(z, s; h) = \sum_{i=1}^{3} r_i^j \Phi(z, s, h + r_i), \qquad j = 0, 1, 2.$$

Since $\theta\Phi(z,s,a) = \Phi(z,s-1,a) - a\Phi(z,s,a)$,

$$\begin{aligned}\theta F_0 &= F_0(z,s-1;h) - hF_0 - F_1,\\ \theta F_1 &= F_1(z,s-1;h) - hF_1 - F_2,\\ \theta F_2 &= F_2(z,s-1;h) - hF_2 - \sigma F_1 + \tau F_0.\end{aligned} \tag{11.4}$$

The last line uses $r_i^3 = \sigma r_i - \tau$. This supplies argument derivatives without differentiating branch choices for individual cubic roots.

Anchored h-antiderivatives of the order jets of $\mathcal{L}$ are obtained by replacing $\mathcal{Z}$ with $\mathcal{L}$ in (10.7)–(10.8). The same removal proof works: at nonpositive integer order the Lerch series is a polynomial in h with coefficients rational in z, and the trace cancels its leading terms. At $z = 1$ one must use the Hurwitz trace version or a justified endpoint limit rather than substituting into an individual negative-order Lerch series.

12 Audit findings and claims that must not be made

12.1 The inspected mixed-resonance proof

No new mathematical error was established in the inspected canonical mixed-resonance chapter. Its centered $x_n = n + a/2$ normalization, the first resonant subtraction, and the sign of the cubic counterterm at $a = 1$ agree with the independent calculation in Section 3. The explicit Bernoulli cancellations are also checked by exact symbolic arithmetic in the supplied code. This is a targeted audit, not an assertion that every statement in the manuscript or incoming archives has been checked.

The necessary integration safeguards are mathematical rather than cosmetic. The $n = 0$ term must be included in the present moments; omitting it changes an unweighted tail-power identity by the corresponding product of zeta values. The weight here is $2n+1$, not $n+1/2$. The roots of the cubic have product $-\tau$, whereas the odd parameter of the Gamma quotient is $+\tau$. At $x = 0$ the complete right side of the digamma formulas must be evaluated by a removable limit. At higher Stieltjes index, the spectral pole must be canceled before taking Taylor coefficients.

12.2 A false extrapolation, not an attributed repository conjecture

The cases $d = 2, 3$ might suggest

$$\sum_n (2n+1)T_3(n)^d \stackrel{?}{=} (2 - 2^{1-d})\zeta(3d-2). \tag{12.1}$$

This is false already at $d = 4$. We do not attribute (12.1) to any existing source. A rational countercheck is enough: positivity gives a lower bound $\left(\sum_{j=1}^4 j^{-3}\right)^4$ for the left side, while

$$\frac{15}{8}\zeta(10) < \frac{15}{8}\left(1 + 2^{-10} + \frac{1}{9\cdot 2^9}\right) < \left(\sum_{j=1}^4 j^{-3}\right)^4.$$

The last inequality is an exact comparison of rational numbers. The correct fourth-power formula is (6.8). This small rejection is included to prevent a misleading simplification, not because inequalities are a research target of this article.

12.3 The Gaussian conjectures are not solved here

The canonical S_4 identity already has an exact octahedral certificate; the inspected discovery chapter distinguishes a rigorously rejected old S_8 vector from the surviving revised candidate [3]. The present descent concerns ordinary odd zeta tails. We have not proved a bridge from the mixed-color Gaussian S_6 or revised S_8 residual to this invariant family. Their status is unchanged by this work. Nor does the appearance of $\zeta(7,3)$ prove any arithmetic irreducibility.

13 Reproducibility and manuscript integration

The analytic proofs above are independent of the numerical computations. The package includes one self-contained LaTeX source, its compiled PDF, verification code, machine-readable results, a coefficient catalog, an integration note, and a compact theorem insert.

13.1 Finite exact checks

The exact suite makes **596 assertions**. It compares two independent constructions of the invariant exponential coefficients through $0 \le a \le 4$, $0 \le d \le 8$; checks their tail weight and the two-factor parity descent; compares the Newton recurrence with the closed formula through degree 30; checks the harmonic/Pochhammer coefficients through $k = 20$ and order five; verifies finite minimum-square sums against ordered partitions using rational arithmetic; checks the four printed low-degree polynomials and the Bernoulli constants; and verifies the negative-order Taylor cancellation for the first eight primitive orders.

The catalog records the 35 nonlinear pairs $0 \le a \le 4$, $2 \le d \le 8$. These tests verify finite algebra relative to the formulas stated in the article. They are not a proof-assistant certification of Gamma continuation, Tonelli's theorem, or normal convergence.

13.2 Independent numerical diagnostics

The numerical program uses defining Hurwitz tails for the quadratic, cubic, quartic and quintic identities. The depth-two coordinate in the quartic formula is evaluated independently by its polylogarithmic integral. The digamma formulas are checked at real points on both sides of a pole and at a nonreal point, always avoiding the pole itself. Further checks compare the full subtracted Gamma master with its digamma value, the invariant Stieltjes expansion with real-argument Stieltjes constants at a repeated-root configuration, anchored primitives with ordinary quadrature, and the finite polylogarithmic primitive with its defining integral.

All **31 numerical checks** passed at **50 decimal working digits**. The output files record working precision, residual, tolerance and pass status for every test actually completed. Numerical tolerances are floating-point diagnostic thresholds, not rigorous error enclosures. In particular, no integer-relation search is used to infer an identity.

13.3 Rebuilding and proposed placement

From the package directory, run

```
python verification/verify.py --exact-only
python verification/verify.py --numeric-only --dps 50
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The proposed report home is the continuation directory

> `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/cubic-invariant-tail-descent/`.

The main insertion follows the mixed-resonance chapter. The trace and primitive results can instead be cross-referenced from the Hurwitz differentiation and integration chapters. All mathematical labels in this article use the prefix `ct:`; the supplied compact insert uses `cti:`. Proposed intake and integration are not claimed to have occurred.

14 Further research questions

1. An elementary certificate for the master identity. Can (2.8) be proved by an explicit finite-difference or contour certificate that avoids the meromorphic continuation of the Rogers–Dougall sum? The cubic product ratio $Q_{n-1}/Q_n = (n^3 - \sigma n + \tau)/(n^3 - \sigma n - \tau)$ is unusually simple. A successful certificate must recover the nonzero constant $1/2 - 2\log 2$, not just its nonlinear coefficients. Such a proof could also provide a shorter analytic foundation for formalization.

2. Full polarization of the symmetric-digamma cube. Theorem 7.1 evaluates the diagonal sum $\sum_n (2n+1)\mathcal{A}_n(x)^3$. Determine a closed formula for $\sum_n (2n+1)\mathcal{A}_n(x)\mathcal{A}_n(y)\mathcal{A}_n(z)$ at independent arguments. Derivatives on the diagonal and the symmetric convolutions proved here do not uniquely determine this three-variable function. A divided-difference formula with controlled collision limits would be particularly useful.

3. The fifth-power normal form. Equation (6.10) provides a small, explicit weight-thirteen starting basket. Reduce its particular combination further by exact shuffle, stuffle and duality relations, keeping a rational certificate. The target is a shorter proven expression for the fifth moment, not a numerical claim that a candidate list is linearly independent.

4. Beyond three roots. For four or more roots with zero sum, the odd part of a Gamma determinant still has a parity filtration, but there are additional independent invariants. Determine which higher hypergeometric summations supply a normally convergent weighted trace identity strong enough to isolate the leading tail moments. Parity of an exponential alone does not supply the missing summation theorem.

5. Colored analogues and the Gaussian obstruction. Develop a roots-of-unity analogue of the cubic product in which the tail indices are restricted to congruence classes or carry colors. A meaningful first target is an exact formula for a colored cubic tail moment whose ordered-partition image contains the mixed-color Gaussian depth-two coordinates. Only a proved transformation of that kind could connect this descent to S_6 or the revised S_8 conjecture.

6. A genuine continuous-order nonlinear deformation. The Lerch trace permits arbitrary order derivatives, whereas the Rogers–Dougall tail descent uses the fixed odd integer grading. Construct a nonlinear master identity with a free spectral order that interpolates the integer moments and whose derivatives produce weighted logarithmic tails. One must not obtain such a deformation by differentiating a family known only at isolated integer orders.

7. Formal verification with analytic hypotheses. The recurrence, Newton algebra, ordered-partition map and primitive removal are finite algebraic targets. A useful next step is to formalize these first, with the normally convergent master identity as an explicit hypothesis, and then formalize the Gamma-continuation and interchange lemmas. The supplied computations already separate those two layers.

Conclusion

The cubic invariant coordinates expose a triangular feature of an existing Gamma resonance: nonlinear coefficients of high tail degree are controlled by lower tail degree, with a two-step parity gap. This yields an exact all-degree descent, particularly simple cubic and quartic evaluations, and meromorphic formulas for symmetric digamma powers. The same invariants organize Hurwitz, Stieltjes, Gamma and Lerch traces without introducing root-branch singularities, and allow every integration constant in the primitive tower to be specified. The result is a concrete continuation of the exact-identity program, with clearly separated classical inputs, proved consequences, numerical diagnostics and remaining conjectures.

References

[1] ProveIt Contributors, *Mixed Gamma derivatives and exact harmonic tail identities*, canonical manuscript chapter `Analysis/Polylogarithms/docs/manuscript/chapters/07-mixed-dougall-identities.tex`, source anchor `87c7db629552afbc2e803001526debe3430a31b0`, inspected 10 October 2026. Repository: `https://github.com/VladimirReshetnikov/ProveIt`.

[2] ProveIt Contributors, *Harmonic Gamma coefficient identities*, canonical chapter `07-harmonic-gamma-coefficients.tex`, same repository and source anchor as [1].

[3] ProveIt Contributors, canonical manuscript `README.md`, `chapters/04-S4-proof.tex` and `chapters/10-discovery.tex`; incoming `README.md` and directory inventory, inspected 10 October 2026. Same repository as [1]. These are source-status references, not assertions of a complete archive audit.

[4] NIST Digital Library of Mathematical Functions, §16.4, especially Eq. 16.4.9, Rogers–Dougall very well-poised sum. `https://dlmf.nist.gov/16.4.E9`. Consulted 10 October 2026.

[5] NIST Digital Library of Mathematical Functions, §5.8, infinite products, and §5.11, asymptotic expansions of Gamma functions and ratios. `https://dlmf.nist.gov/5.8`; `https://dlmf.nist.gov/5.11`. Consulted 10 October 2026.

[6] NIST Digital Library of Mathematical Functions, §25.11, Hurwitz zeta function: shift recurrence, parameter derivatives, negative-integer values and $\zeta'(0,h)$; §25.14, Lerch's transcendent. `https://dlmf.nist.gov/25.11`; `https://dlmf.nist.gov/25.14`. Consulted 10 October 2026.

[7] Michael E. Hoffman, *Sums of Products of Riemann Zeta Tails*, arXiv:1602.03399, 2016. In particular §2 for the Euler depth-two identities and §§3–4 for products of tails. `https://arxiv.org/abs/1602.03399`.